

Test 1 Handout Sheet

Given a recurrence of the form

$$T(n) = aT(n/b) + f(n)$$

where a and b are constants with $a \geq 1$ and $b > 1$.

Three cases must be considered.

Case 1: If

$$f(n) = O(n^{\log_b(a) - \epsilon})$$

where $\epsilon > 0$, then

$$T(n) = \Theta(n^{\log_b(a)})$$

Case 2: If

$$f(n) = \Theta(n^{\log_b(a)})$$

then

$$T(n) = \Theta(n^{\log_b(a)} \lg n)$$

Case 3: If

$$f(n) = \Omega(n^{\log_b(a) + \epsilon})$$

where $\epsilon > 0$ and $af(n/b) \leq c f(n)$ for some $c < 1$ then

$$T(n) = \Theta(f(n))$$

- for this case you must show $af(n/b) \leq c f(n)$ for some $c < 1$

Some Useful Summation Formulae

$$\sum_{i=1}^n 1 = n$$

$$\sum_{i=1}^n C = n * C$$

$$\sum_{i=1}^n i = \frac{n * (n + 1)}{2}$$

$$\sum_{i=0}^n (n - i) = \sum_{i=0}^n i$$

$$\sum_{i=1}^n i^2 = \frac{n * (n + 1) * (2n + 1)}{6}$$

$$\sum_{i=0}^n 2^i = 2^{n+1} - 1$$

$$\sum_{i=0}^n C^i = \frac{C^{n+1} - 1}{C - 1}$$

$$\sum_{i=1}^n i * 2^i = (n - 1) * 2^{n+1} + 2$$

$$\sum_{i=1}^n \frac{1}{i} \approx \ln n$$

$$\sum_{i=1}^n \lg i \approx n * \lg n - 1.5$$